

Lab03_FabianoJunior

#1

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
##1
getStats <- function(x, pos) {
  x = x%>%
    filter(!startsWith(DESTINATION_AIRPORT, "1")) %>%
    filter(complete.cases()) %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      qtd = n()
    )
  x
}
flights_col = cols_only(DESTINATION_AIRPORT = 'c', ARRIVAL_DELAY = 'd')
flights = read_csv_chunked("flights.csv",
```

```

        callback = DataFrameCallback$new(getStats),
        chunk_size = 1000000,
        col_types = flights_col)
airlines = read_csv("airlines.csv")

```

Rows: 14 Columns: 2

```

-- Column specification -----
Delimiter: ","
chr (2): IATA_CODE, AIRLINE

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

airports = read_csv("airports.csv",
                    col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))

```

Rows: 322 Columns: 5

```

-- Column specification -----
Delimiter: ","
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```

flights = bind_rows(flights) %>%
  group_by(DESTINATION_AIRPORT)%>%
  summarise(
    soma_atrasos = sum(soma_atrasos),
    qtd = sum(qtd),
    media_atraso = soma_atrasos / qtd
  )

```

#2

```

##2
### a coluna chave em flights eh DESTINATION_AIRPORT
### a coluna chave em CITY e STATE
### aeroportos que nao tem em airports mas tem em flights

```

```

ausentes <- flights %>%
  anti_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )
### todos aeroportos estao listados

flights <- left_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)

```

#3

```

airports %>%
  group_by(STATE) %>%
  summarise(qtd_aeroportos = n()) %>%
  arrange(desc(qtd_aeroportos))

```

```

# A tibble: 54 x 2
  STATE qtd_aeroportos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC             8
# i 44 more rows

```

#4

```

library(leaflet)
this = leaflet(flights) %>%
  addTiles()%>%
  addMarkers(

```

```
lng = ~LONGITUDE,  
lat = ~LATITUDE,  
popup = ~paste(  
  "Aeroporto:", DESTINATION_AIRPORT,  
  "<br>Atraso médio:", media_atraso  
)  
this
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 02:15:44 -03"
```